Liver physiology

Hawler medical university

College of medicine

Module: GIT

BY

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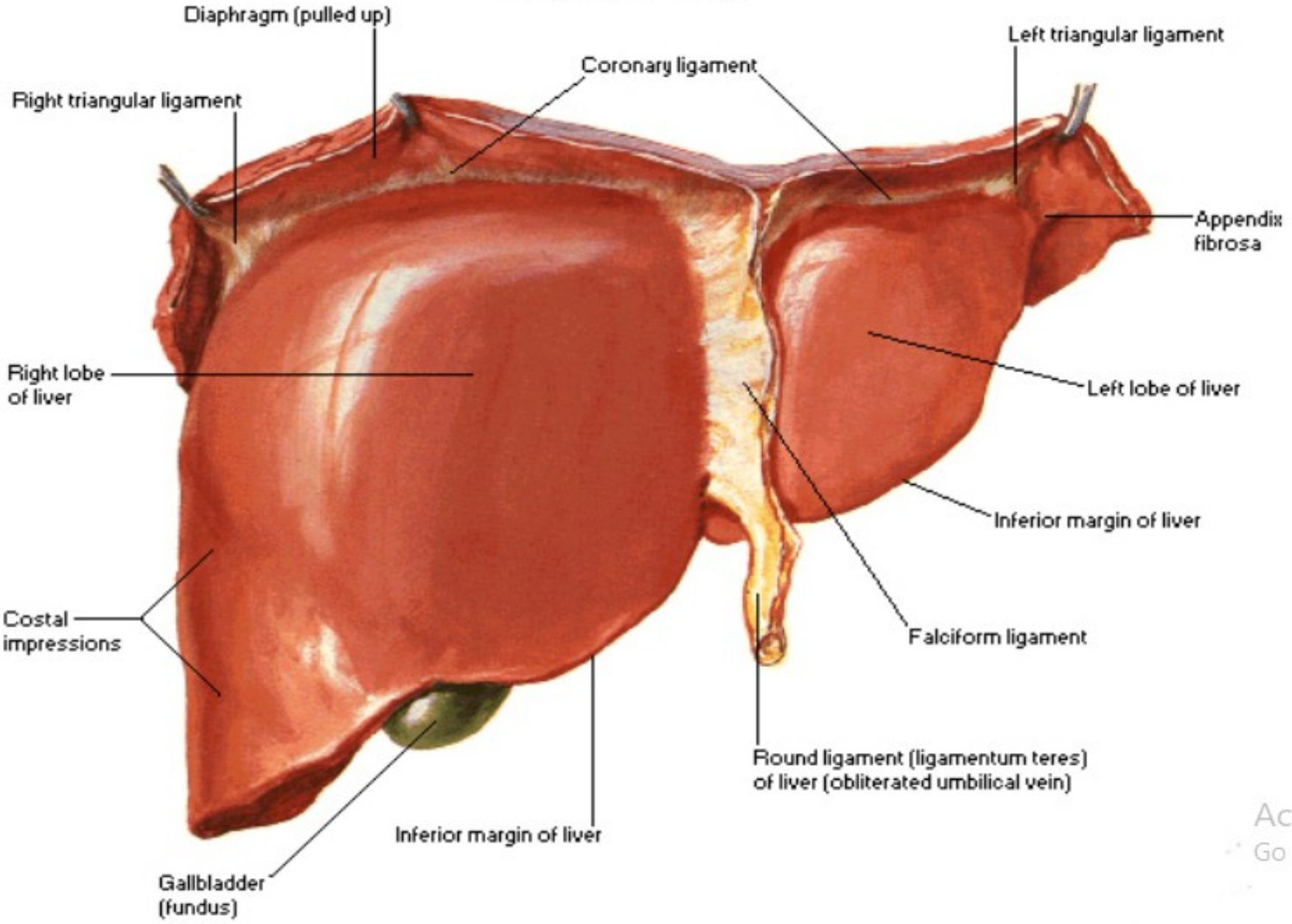
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Introduction

- The liver is the largest internal organ and gland. It is second only to the skin in size.
- Can perform a multi-task commands at once.
- Is divided into two principal lobes (large right and smaller left).
- According to blood vessel distribution, the left lobe is further include quadrate and caudate lobes.
- each lobe consist of hepatic lobules.
- Lobules are the functional units of liver.
- Human liver contains 50.000 to 100.000 individual lobules.

Surfaces and Bed of Liver

Anterior View



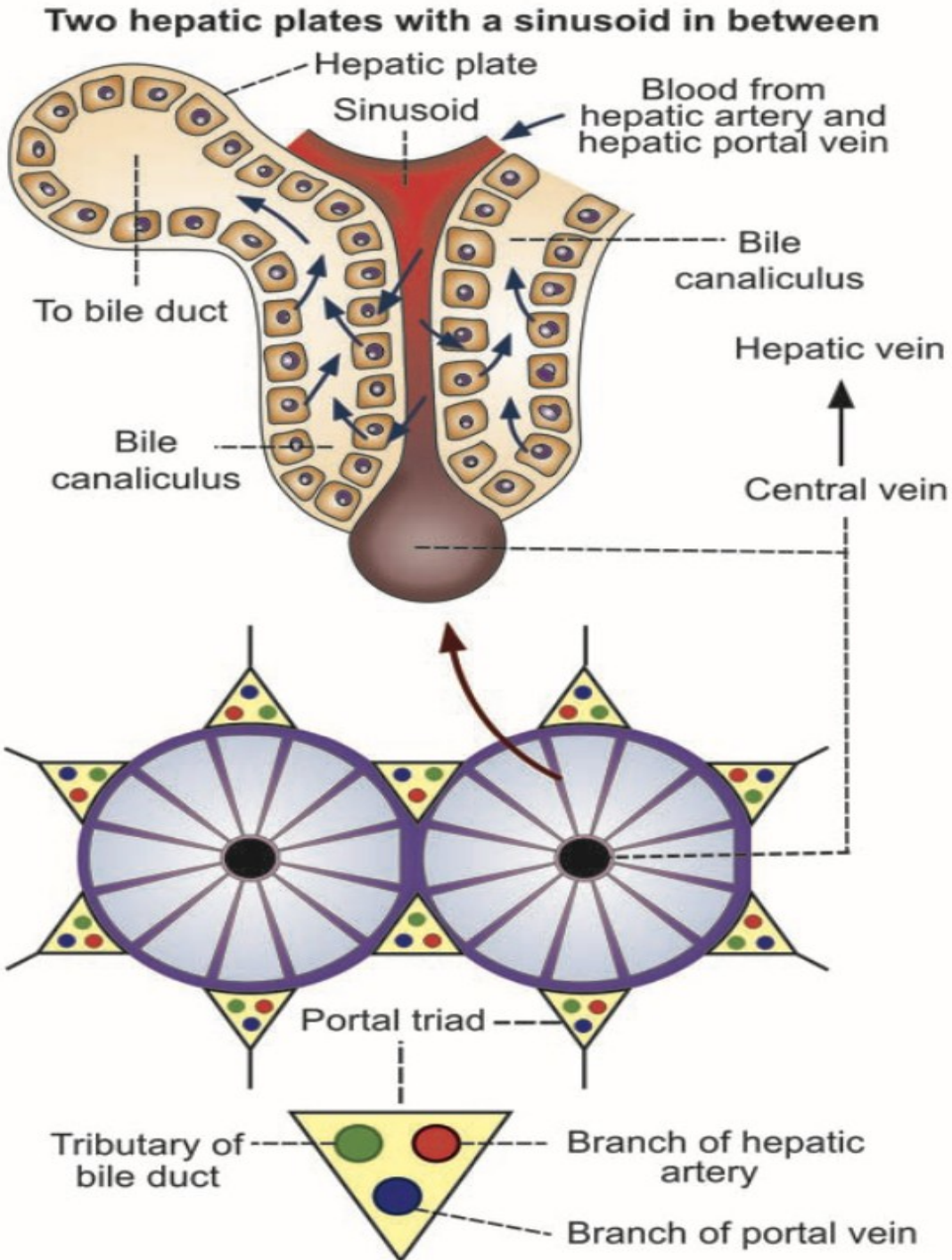
- Lobules are consist of hepatic cells called hepatocytes, (up to 30% of hepatocytes are double nucleated).
- Hepatocytes form a hepatic plates (that are 1-2 cells thick secretory acinar cells) arranged in columns.
- between the two columns of each plate lies a bile canaliculus. (bile ductulus).
- Bile ducts and bile canaliculus are lined by a polarized epithelial cells called cholangiocytes, that modify and transport bile generations.

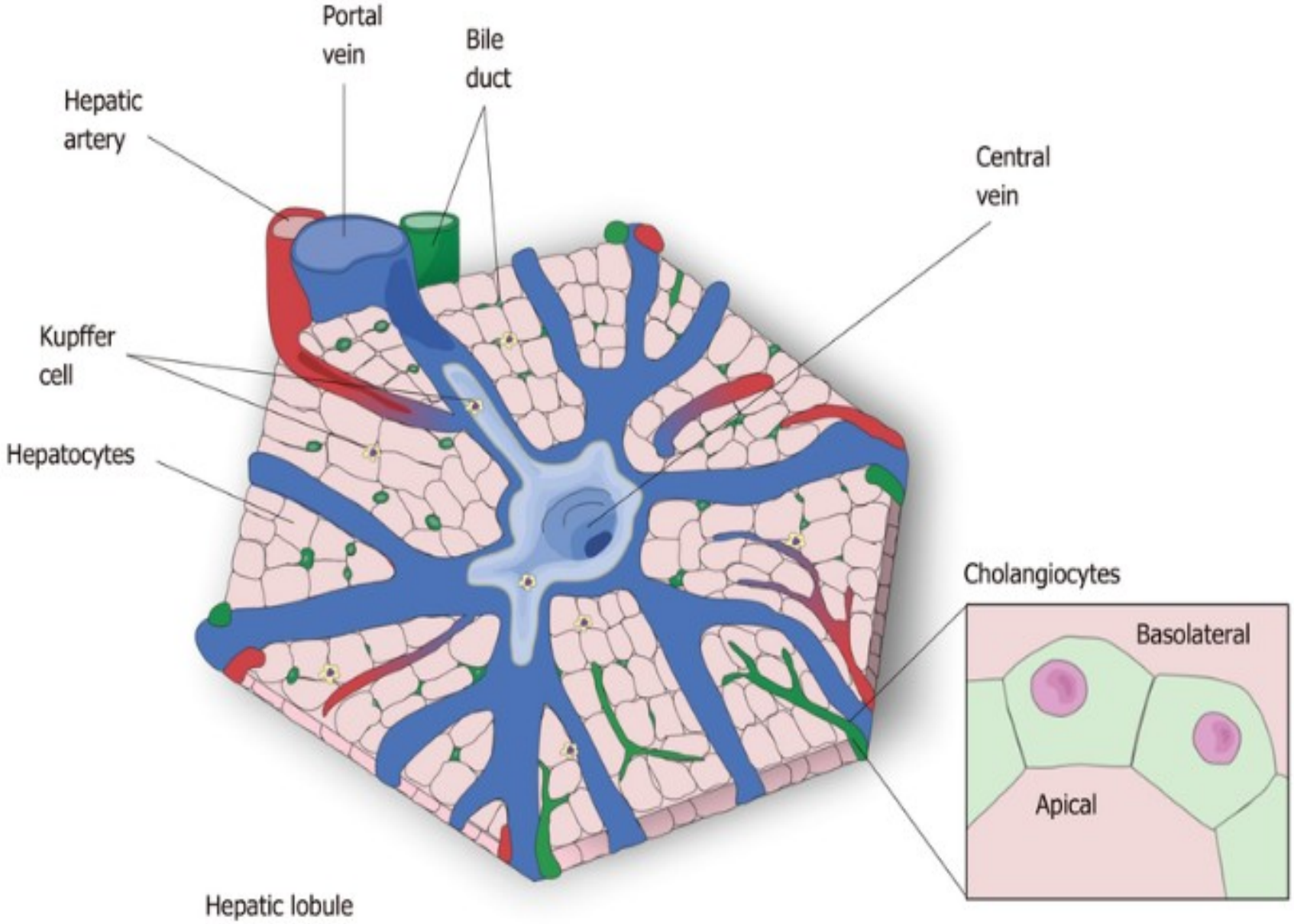
- The plates are separated from each other by large capillary spaces called sinusoids.
- sinusoids are lined by endothelial cells with flattened processes and fenestrae—openings 150 to 175 nanometers in diameter that make the sinusoids very porous.
- The sinusoids also contain phagocytic Kupffer cells.

- Kupffer cells are effective particulate filtration system that almost none of the bacteria from the gastrointestinal tract pass from the portal blood into the general systemic circulation.

Liver Lobule

- The lobule is a **honeycomb-like structure (hexagonal)**
- In the middle of each lobule is a central vein, and at the periphery of each lobule are (portal triads) which consist of **branches of three vessels (hepatic artery, portal vein, and bile ducts)** which open into the sinusoids between hepatic plates.
- Arterial blood and portal venous blood, containing molecules absorbed in the GI tract, thus mix as the blood flows within the sinusoids from the periphery of the lobule to the central vein.
- The central veins of different liver lobules converge to form the hepatic vein, which carries blood from the liver to the inferior vena cava.
- The average transit time for blood across the liver lobule from the portal venule to the central hepatic vein is about 8.3 second


Two hepatic plates with a sinusoid in between
Hepatic plate
Sinusoid
Blood from hepatic artery and hepatic portal vein
Bile canaliculus
To bile duct
Hepatic vein
Bile canaliculus
Central vein
Portal triad
Tributary of bile duct
Branch of hepatic artery
Branch of portal vein


Portal vein
Bile duct
Hepatic artery
Central vein
Kupffer cell
Hepatocytes
Cholangiocytes
Basolateral
Apical
Hepatic lobule

Hiepatic blood supply

- Total hepatic blood flow is approximately **1.2–1.5 L/min in a resting adult**
- The liver has dual blood supply from:
- 1- Hepatic artery provides 25% of cardiac output (branch of coeliac trunk artery from abdominal aorta).
- The hepatic artery is oxygenated nutrient poor blood enters the liver through two branches.
- Each branch enters a portal triad.

Hepatic blood supply

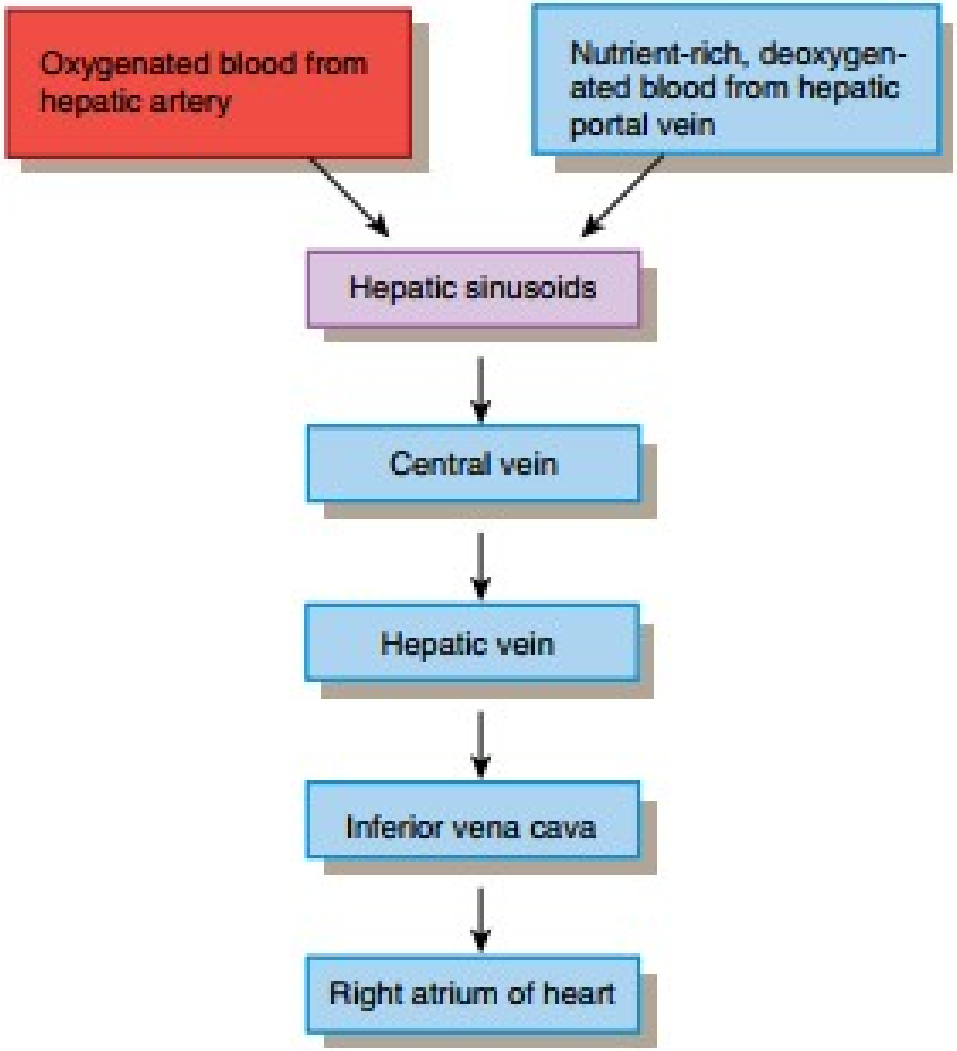
- 2- Portal vein from mesenteric and splenic veins (supplies 75% of blood to the liver).
- It brings ***deoxygenated nutrient-rich blood*** from stomach, intestine, spleen and pancreas.
- Portal blood is rich in monosaccharides and amino acids. It also contains bile salts, bilirubin, urobilinogen and GI hormones.
- Although it has already passed through another capillary bed, portal blood still contains oxygen and makes a substantial contribution to hepatic oxygen delivery. The hepatic artery contributes a smaller blood volume with a higher oxygen concentration.

Note/ Flow of blood from intestine to liver through portal vein is known as enterohepatic circulation.

Within the liver, arterial and portal venous branches become progressively smaller, and their blood ultimately mixes in the hepatic sinusoids

- Hepatic cells obtain oxygen and nutrients from the sinusoid, where oxygen, most of the nutrients, and certain toxic substances are taken up by the hepatocytes.

The liver receives oxygenated blood via the hepatic artery and nutrient-rich deoxygenated blood via the hepatic portal vein.



Hepatic blood supply

- Products manufactured by the hepatocytes and nutrients needed by other cells are secreted back into the central veins, larger collecting veins, and the hepatic veins before entering the inferior vena cava.

- The percentage of blood supplementation to liver (25% from hepatic artery+ 75% from portal vein) is shifted to 90% of blood delivered to liver from portal vein during post prandial effect.

Hemodynamic feedback loops

- An important protective mechanism is the **hepatic arterial buffer response**.
- When portal venous flow decreases, hepatic arterial resistance falls and arterial flow increases, partially compensating for the reduction.
- A major explanation is that reduced portal flow decreases the washout of locally produced adenosine, allowing it to accumulate and dilate hepatic arterioles. This response helps stabilize hepatic oxygen delivery, but it does not completely preserve total liver blood flow and cannot be assumed to protect the biliary tree from loss of its arterial supply.

Blood pressure dynamics

- Sinusoidal equalizer:
- The two systems (75% portal vein, low pressure 9 mmHg and 25% hepatic artery high pressure 90 mmHg) merge together within highly porous, fenestrated sinusoids. The structural design causes a major drop of pressure, maintaining of an intra-sinusoidal pressure of only 4 mmHg.
- Blood safely flows to vena cava at 0 mmHg.

Why low pressure is essential

- 1- to prevent massive fluid leakage (Ascites).
- Because plasma proteins across the large vessel holes freely, the protective osmotic pressure that normally keeps water inside blood vessels is eliminated. This leaves hydrostatic pressure as the single force controlling fluid movement.
- 2- maximizing nutrient and oxygen exchange.

Ascites = Fluid Transudation in to Abdominal Cavity

- High hepatic vascular pressure (3-7 mmHg above normal), causes excessive amounts of fluid to transude into the lymph and leak through the outer surface of the liver capsule directly into the abdominal cavity.
- This fluid is almost pure plasma, containing 80-90% as much protein as normal plasma.
- At vena caval pressures of 10 to 15 mm Hg, hepatic lymph flow increases to as much as 20 times normal, and the “sweating” from the surface of liver increased and causes large amounts of free fluid in the abdominal cavity, which is called Ascites
- Blockage of portal flow through the liver also causes high capillary pressures in the entire portal vascular system of the gastrointestinal tract, resulting in edema of the gut wall and transudation of fluid through the serosa of the gut into the abdominal cavity. This, too, can causes ascites.

Mechanism/ decreased hepatic albumin synthesis leads to hypoalbuminemia and decreased plasma oncotic pressure.

The need for dual circulation

- Not all substances absorbed from the gastrointestinal tract can be used in their absorbed form by the cells.
- The liver changes the chemical compositions of many of these substances to more usable forms, and other tissues of the body—fat cells, gastrointestinal mucosa, kidneys, and endocrine glands—help modify the absorbed substances or store them until they are needed.
- The liver also eliminates certain waste products produced in the body and toxic substances that are ingested.

Bile synthesis

- Bile production begins in hepatocytes and continues through modification by **cholangiocytes**.
- The liver produces roughly **0.5–1 L of bile daily**
- The major constituents of bile are bile pigment (bilirubin), bile salts, phospholipids (mainly lecithin), cholesterol, and inorganic ions.
- Bile is produced by the hepatocytes and secreted into thin channels called bile canaliculi, located within each hepatic plate.
- Because blood travels in the sinusoids and bile travels in the opposite direction within the hepatic plates, blood and bile do not mix in the liver lobules.
- These bile canaliculi are drained into interlobular bile ducts, and these coalesce via interlobular bile duct to form the right and left hepatic ducts, these ducts join outside the liver to form the common hepatic duct.
- The cystic duct join the gallbladder. The hepatic duct unites with the cystic duct to form the common bile duct. Then drains to the doudenem by hepatopancreatic apmulla (Ampulla of Vater).

- The ductules (canaliculi) scavenge plasma constituents, such as glucose and amino acids, and return them to the circulation by active transport.
- Glutathione is also hydrolyzed to its constituent amino acids by an enzyme, gamma glutamyltranspeptidase (GGT), expressed on the apical membrane of the cholangiocytes.
- Removal of glucose and amino acids is likely important to prevent bacterial overgrowth in the bile, particularly during gallbladder storage.
- The ductules also secrete bicarbonate in response to secretin in the postprandial period, as well as IgA and mucus for protection

Content of bile

- The principal bile pigment is bilirubin.
- The phagocytosis of aged red blood cells liberates iron, globin, and bilirubin (derived from heme).
- The iron and globin are recycled; the bilirubin is secreted into the bile and is eventually broken down in the intestine.
- One of its breakdown products—stercobilin—gives feces their normal brown color.
- Bile salts, which are sodium salts and potassium salts of bile acids (mostly chenodeoxycholic acid and cholic acid), play a role in emulsification, the breakdown of large lipid globules into a suspension of small lipid globules.
- The small lipid globules present a very large surface area that allows pancreatic lipase to more rapidly accomplish digestion of triglycerides.

Biliary system

- The component of biliary system are the liver, gallbladder and bile duct, duodenum, ileum, and portal circulation.
- Bile flows out of the liver to the bile duct and fills the gallbladder and stored.
- Gallbladder then concentrate the bile salts by absorption of water and ions.
- When chyme reaches the small intestine, CCK is secreted.
- In small intestine the bile salts emulsify and solubilize lipids
- When lipid absorption is complete, the bile salts re-circulated to the liver by enterohepatic circulation (from ileum to the portal system)
- Extraction of bile salts from the portal blood by hepatocytes.

Note/ The recirculation of bile salts to the liver reduces the demand to synthesis new bile salt.

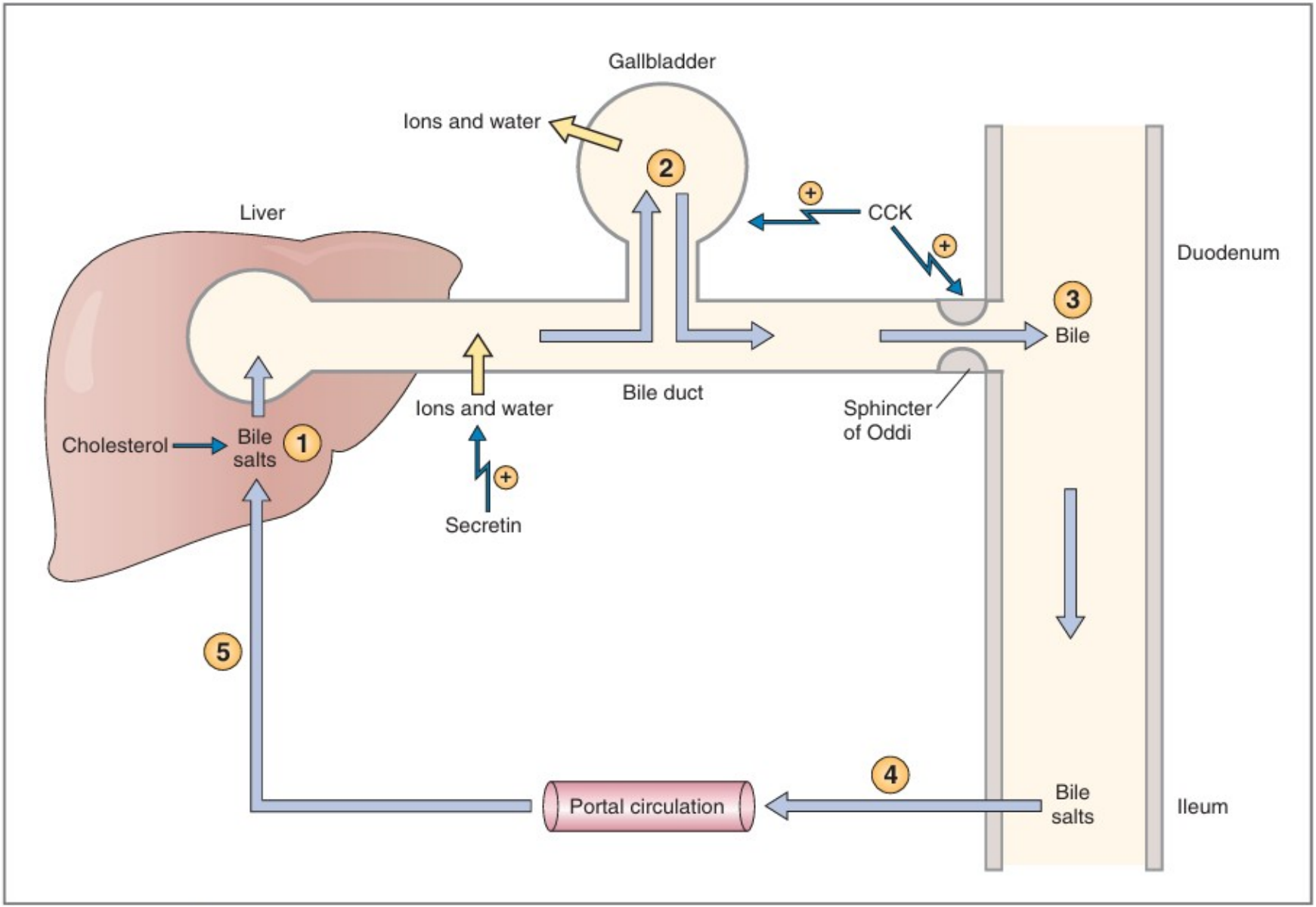


Fig. 8.24 Secretion and enterohepatic circulation of bile salts. *Light blue arrows* show the path of bile flow; *yellow arrows* show the movement of ions and water. *CCK,* Cholecystokinin.

Cholycystokinin CCK and its role in emptying the gallbladder

is a classic gut hormone that has been known for almost a century to regulate gallbladder emptying, pancreatic enzyme secretion, and gastrointestinal motor activity.

released by intestinal I cells in response to digestion products, particularly fats and proteins

Has two separate but coordinated function:

1- it stimulate contraction of the gallbladder(reduces resistance at the sphincter of Oddi, allowing bile to pass through the cystic duct and common bile duct into the duodenum by ampula of vatar

2- Relaxation of sphincter of Oddi. Causing the stored bile to flow from gallbladder into the lumen of the duodenum.

REGULATION OF BILIARY SECRETION

- When food enters the mouth, the resistance of the sphincter of Oddi decreases under both neural and hormonal influences.
- Fatty acids and amino acids in the duodenum release CCK, which causes gallbladder contraction.
- The production of bile is increased by stimulation of the vagus nerves and by the hormone secretin, which increases the water and HCO_3^- content of bile.
- Substances that increase the secretion of bile are known as choleretics. Bile acids them selves are among the most important physiologic choleretics.

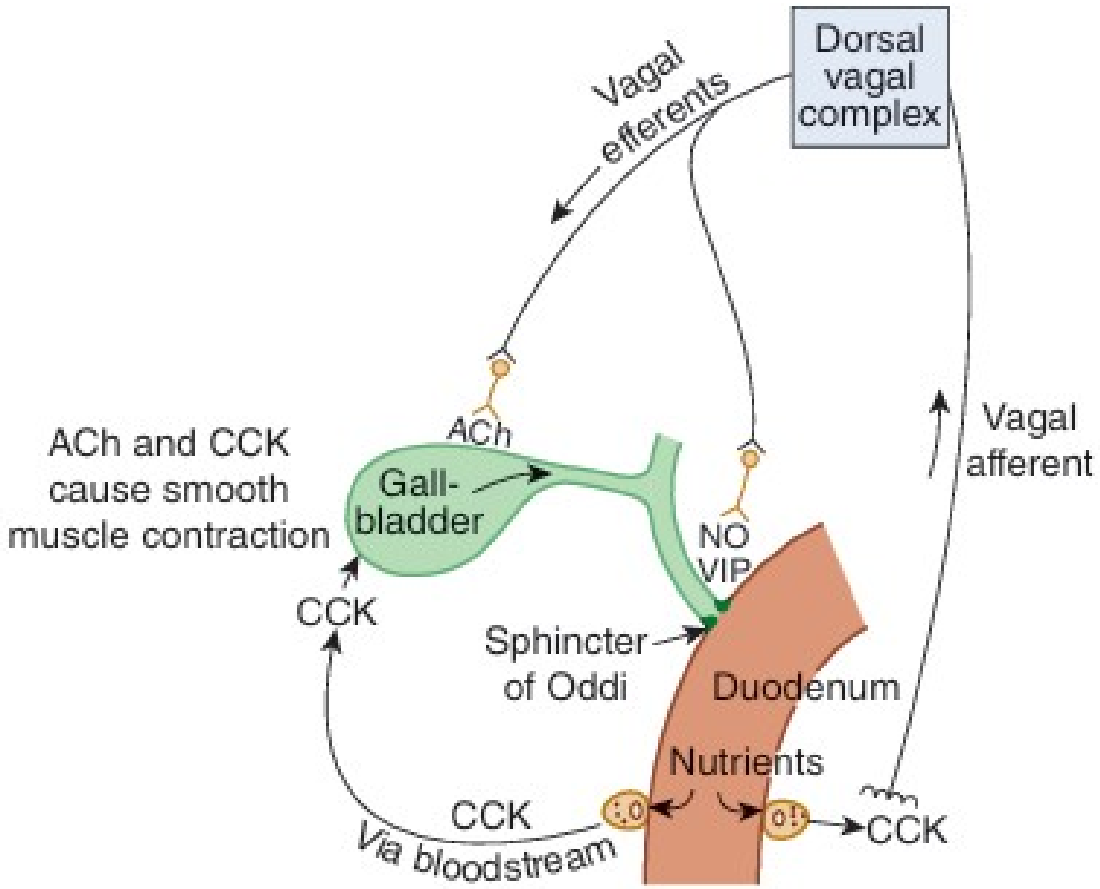


FIGURE 28–8 Neurohumoral control of gallbladder contraction and biliary secretion. Endocrine release of cholecystokinin (CCK) in response to nutrients causes gallbladder contraction. CCK also activates vagal afferents to trigger a vagovagal reflex that reinforces gallbladder contraction (via acetylcholine [ACh]) and relaxation of the sphincter of Oddi to permit bile outflow (via NO and vasoactive intestinal polypeptide [VIP]).

Liver as reservoir of blood

The liver is a large, expandable, venous organ capable of acting as a valuable blood reservoir in times of excess blood volume and capable of supplying extra blood in times of diminished blood volume.

- Large quantities of blood can be stored in its blood vessels (10% of the body's total blood volume).
- Up to 0.5 to 1 liter of extra blood is occasionally stored in the hepatic veins and sinuses in cases of high right atrial pressure (cardiac failure).

lymph formation and drainage

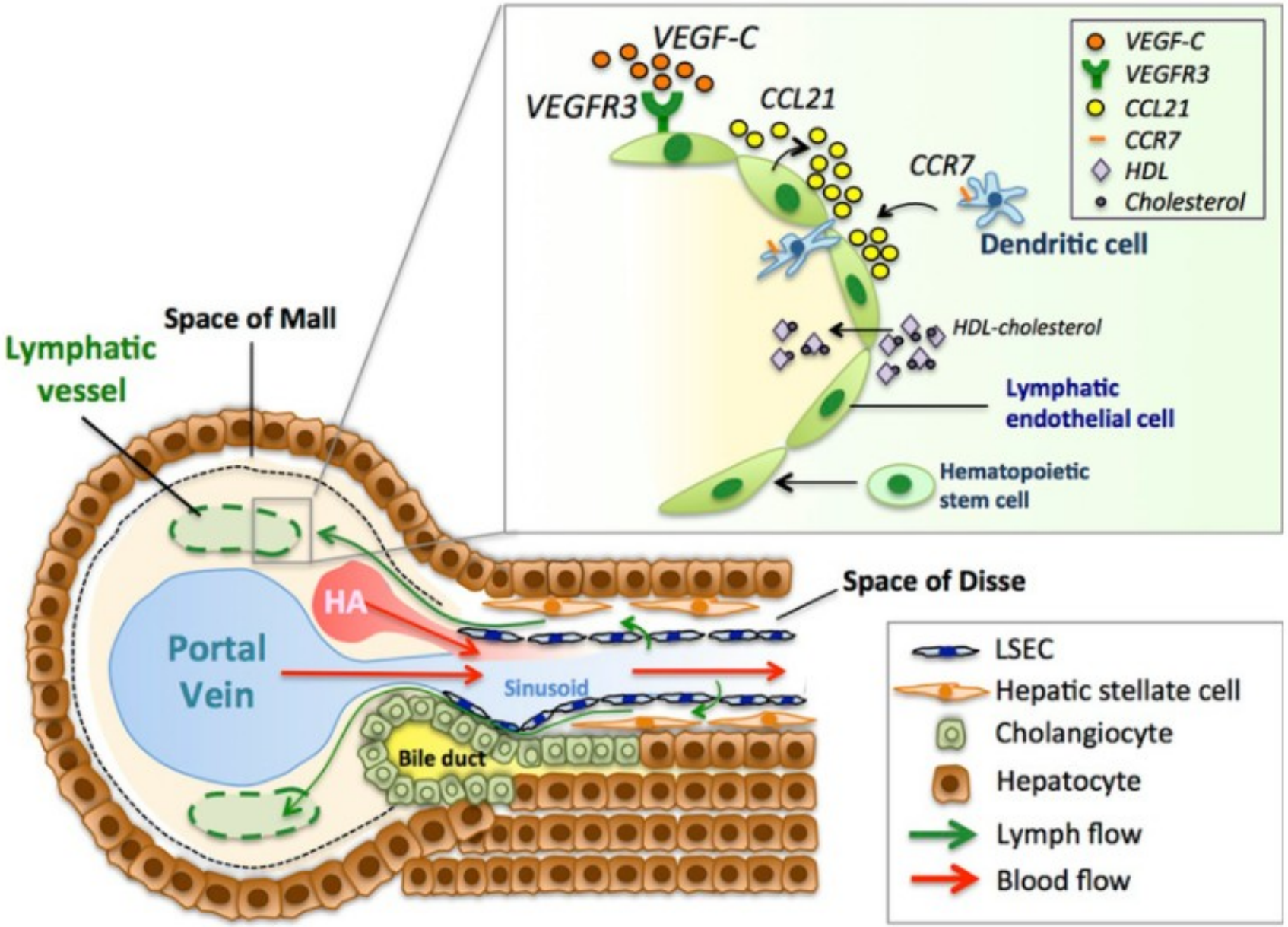
- The liver is the largest lymph producing organ, accounting for 25%–50% of lymph passing through the thoracic duct.
- The hepatic lymphatic system helps to remove waste products and immune cells derived from the sinusoidal microcirculation as well as hepatocytes and non-parenchymal cells in the form of lymph, by transporting lymph through lymphatic vessels to draining lymph nodes.

- A majority of lymphatic vessels are located in the portal tract, which accommodates the portal vein, the hepatic artery and bile ducts.
- Sinusoids are covered by fenestrated liver sinusoidal endothelial cells (LSECs).
- Hepatic lymphatic fluid originates mainly from plasma components filtered through fenestrae of LSECs and flows into the space of Disse, the interstitial space between LSECs and hepatocytes.
- Hepatic stellate cells are also located in this space.
- Lymphatic fluid in the space of Disse primarily flows through the space of Mall, a space between the stroma of the portal tract and the outermost hepatocytes, into the interstitium of the portal tract and then into lymphatic vessels.

- The deep system drains most of the hepatic parenchyma and follows the portal tracts and hepatic veins; lymph accompanying the portal triads converges toward the porta hepatis, passes through hepatic and regional lymph nodes, and subsequently reaches the celiac and para-aortic lymphatic pathways, ultimately communicating with the cisterna chyli.
- Lymphatic vessels associated with the hepatic veins and inferior vena cava, as well as superficial capsular lymphatics, provide additional drainage routes toward diaphragmatic and mediastinal lymph nodes and may communicate directly with the thoracic duct.
- From the cisterna chyli, hepatic lymph enters the thoracic duct, which ascends through the posterior mediastinum and returns the lymph to the systemic venous circulation at the left venous angle, the junction of the left internal jugular and left subclavian veins.

- Thus, the hepatic lymphatic pathway can be summarized as hepatic sinusoids → space of Disse → space of Mall/portal lymphatics → hepatic and regional lymph nodes → celiac/para-aortic pathways → cisterna chyli → thoracic duct → left venous angle

- Hepatic lymphatic vessels are connected to one or more draining lymph nodes outside the liver.
- Antigen-presenting cells (APCs) including dendritic cells and macrophages in lymph interact with lymphocytes in draining lymph nodes, facilitating adaptive immune responses


VEGF-C
VEGFR3
CCL21
CCR7
Dendritic cell
HDL-cholesterol
Lymphatic endothelial cell
Hematopoietic stem cell
VEGF-C
VEGFR3
CCL21
CCR7
HDL
Cholesterol
Space of Mall
Lymphatic vessel
Space of Disse
HA
Portal Vein
Sinusoid
Bile duct
LSEC
Hepatic stellate cell
Cholangiocyte
Hepatocyte
Lymph flow
Blood flow

Functions of the Liver and Gallbladder

1- Synthesis and transportation of bile. As described earlier in this lecture).

2- Bilirubin metabolism and excretion:

Most of the bilirubin in the body is formed in the tissues by the breakdown of hemoglobin. The bilirubin is bound to albumin in the circulation.

Most of it is tightly bound, but some of it can dissociate in the liver, and free bilirubin enters liver cells via a member of the organic anion transporting polypeptide (OATP) family, and then becomes bound to cytoplasmic proteins.

It is next conjugated to glucuronic acid in a reaction catalyzed by the enzyme glucuronyl transferase (UDP-glucuronosyltransferase).

This enzyme is located primarily in the smooth endoplasmic reticu lum.

Each bilirubin molecule reacts with two uridine diphos phoglucuronic acid (UDPGA) molecules to form bilirubin diglucuronide.

This glucuronide, which is more water-soluble than the free bilirubin, is then transported against a concentration gradient most likely by an active transporter known as multidrug resistance protein-2 (MRP-2) into the bile canaliculi.

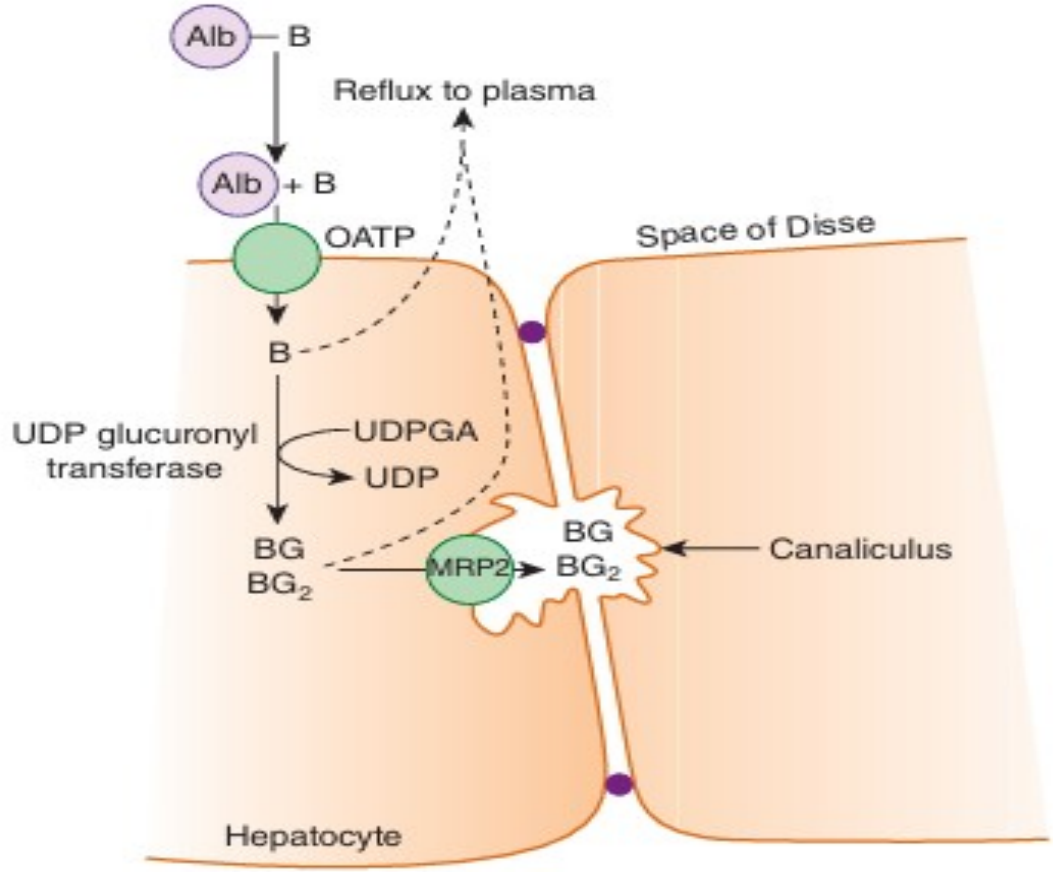


FIGURE 28-6 **Handling of bilirubin by hepatocytes.** Albumin (Alb)-bound bilirubin (B) enters the space of Disse adjacent to the basolateral membrane of hepatocytes, and bilirubin is selectively transported into the hepatocyte. Here, it is conjugated with glucuronic acid (G). The conjugates are secreted into bile via the multidrug resistance protein 2 (MRP-2). Some unconjugated and conjugated bilirubin also refluxes into the plasma. The purple circles linking the two adjacent cells represent the tight junctions. BG, bilirubin monoglucuronide; BG_2, bilirubin diglucuronide; OATP, organic anion transporting polypeptide.

Functions of the Liver and Gallbladder

- 3- **SYNTHESIS OF PLASMA PROTEINS**
- acute-phase proteins, proteins synthesized and secreted into the plasma on exposure to stressful stimuli.
- Others are proteins that transport steroids and other hormones in the plasma, and still others are clotting factors.
- The only major class of plasma proteins not synthesized by the liver is the immunoglobulins.

TABLE 28–1 Principal functions of the liver.

Formation and secretion of bile
Nutrient and vitamin metabolism
Glucose and other sugars
Amino acids
Lipids
Fatty acids
Cholesterol
Lipoproteins
Fat-soluble vitamins
Water-soluble vitamins
Inactivation of various substances
Toxins
Steroids
Other hormones
Synthesis of plasma proteins
Acute-phase proteins
Albumin
Clotting factors
Steroid-binding and other hormone-binding proteins
Immunity
Kupffer cells

Functions of the Liver and Gallbladder

- 4- **Ammonia metabolism and secretion:**
- Ammonia levels must be carefully con trolled because it is toxic to the central nervous system (CNS), and freely permeable across the blood-brain barrier.
- The liver is the only organ in which the complete urea cycle (also known as the Krebs-Henseleit cycle) is expressed.
- This converts circulating ammonia to urea, which can then be excreted in the urine.
- Ammonia in the circulation comes primarily from the colon and kidneys with lesser amounts deriving from the break down of red blood cells and from metabolism in the muscles.

Functions of the Liver and Gallbladder

- 5- **Detoxification**
- The liver detoxifies the blood of substances originating from the gut or elsewhere in the body.
- Part of this function is physical: bacteria and other particulates are trapped in and broken down by the Kupffer cells.
- The remaining reactions are biochemical, and mediated in their first stages by the large number of cytochrome P450 enzymes expressed in hepatocytes.
- These convert xenobiotics and other toxins to inactive, less lipophilic metabolites.

Functions of the Liver and Gallbladder

- Detoxification reactions are divided into:
- 1- phase I (oxidation, hydroxylation, and other reactions mediated by cytochrome P450s).
- 2- phase II (esterification).

Functions of the Liver and Gallbladder

7- METABOLIC FUNCTION: It is the organ where maximum metabolic reactions such as metabolism of carbohydrates, proteins, fats, vitamins and many hormones are carried out.

8- STORAGE FUNCTION: Many substances like glycogen, amino acids, iron, folic acid and vitamins A, B12 and D are stored in liver.

9- SYNTHETIC FUNCTION:

a. Liver produces glucose by gluconeogenesis. It is the only organ which contain glucose-6-phosphatase, so It is the only organ, which can liberate glucose from glycogen into the circulation. Other organs, e.g. muscle, can also store glycogen but are only able to utilize it locally.

b. It synthesizes all the plasma proteins and other proteins (except immunoglobulins) such as clotting factors, complement factors and hormone binding proteins.

c. It also synthesizes steroids, somatomedin and heparin.

Functions of the Liver and Gallbladder

10. EXCRETORY FUNCTION: Liver excretes cholesterol, bile pigments, heavy metals (like lead, arsenic and bismuth), toxins, bacteria and virus (like that of yellow fever) through bile.

11. HEAT PRODUCTION: Enormous amount of heat is produced in the liver because of metabolic reactions. Liver is the organ where maximum heat is

produced.

12. HEMOPOIETIC FUNCTION:

a. In fetus (hepatic stage), liver produces the blood cells.

b. It stores vit.B12 necessary for erythropoiesis and iron necessary for synthesis of hemoglobin.

c. It produces thrombopoietin that promotes production of thrombocytes.

13. HEMOLYTIC FUNCTION: The senile RBCs after a lifespan of 120 days are destroyed by reticuloendothelial cells (Kupffer cells) of liver.

14. INACTIVATION OF HORMONES AND DRUGS:

a. It catabolizes the hormones such as growth hormone, parathormone, cortisol, insulin, glucagon and estrogen.

b. It also inactivates the drugs, particularly the fat soluble drugs. The fat soluble drugs are converted into water soluble substances, which are excreted through bile or urine.

15. DEFENSIVE FUNCTIONS

a. Reticuloendothelial cells (Kupffer cells) of the liver play an important role in the defense of the body.

1. Foreign bodies such as bacteria or antigens are swallowed and digested by reticuloendothelial cells of liver by means of **phagocytosis.**
2. Reticuloendothelial cells of liver also produce substances like interleukins and tumor necrosis factors, which activate the immune system of the body.

b. Liver is also involved in the detoxification of the foreign bodies (the removal of toxic property of various harmful substances).

Thank you